

Vinay Kapakayala

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EDUCATION

University of Wisconsin–Madison

Master of Science in Physics–Quantum Computing

Relevant Coursework: Quantum System Architecture – Quantum Algorithms – Quantum Computing Lab – Advanced Quantum Computing – Machine Learning with Physics – Intro to Quantum Computing and Quantum Mechanics

Madison, WI

Sep. 2023 – May 2025

Sathyabama University

Bachelor of Engineering in Computer Science

Chennai, IND

Aug. 2018 – May 2022

EXPERIENCE

Quantum Software Engineer – Intern

June 2025 – Present

qBraid

Chicago, IL

- Implemented **OpenPulse** support in **PyQASM**, including pulse-level semantic analysis and validation
- Added Duration tracking, Frame-Collision detection and Qubit Register Consolidation for hardware compatibility
- Enhanced compiler reliability and usability by developing unrolling, parsing, and error-handling features, and integrating CI/CD testing and documentation pipelines

Graduate Researcher – Dr. Matthew Otten

Feb. 2024 – May 2025

University of Wisconsin–Madison

Madison, WI

- Research focuses on finding resource estimates for molecular hamiltonian simulations on quantum computer
- Developing quantum algorithms using Trotterization and Qubitization Quantum phase estimation methods and factoring Hamiltonians with Double factorization and Tensor hypercontraction techniques
- Built using Python and Qualtran, estimated Logical qubits, Toffoli counts and Runtime using Qsharp Estimator

Software Engineer – I

Feb. 2022 – July 2023

FIS Global

Chennai, IND

- Built a C++ module in DataNavigator platform to generate daily PIN denial reports based on defined threshold
- Propose and optimized code with parallel processing to generate reports only on certain conditions
- Designed new templates for daily and monthly reports and tested functionality using MySQL
- Followed Agile methodology and used Jira to manage development, testing, and deployment lifecycle

PROJECTS

WISQ Compiler Enhancement | Python, Qiskit, Qsharp

Apr. 2025

- Developed verification and resource estimation modules to assess circuit correctness and hardware costs
- Verified functional equivalence using Hellinger fidelity and quantified error rates under Qiskit noise models
- Estimated resources via Qsharp-Estimator, highlighting differences in T gates and 2-Qubit gates

Neutral Atom Simulator | Python, Qiskit, Qutip

Dec. 2024

- Created Neutral atom simulator using Qutip and integrated with Qiskit pulse simulator focusing on pulse-level design for RX, RZ and CZ gates using Lindblad master equation and operators
- Implemented customized fidelity functions for gate testing and compared with Qiskit standard gates fidelities
- Achieved 99.999% fidelity for single atom gates and 90-95% fidelity for CZ gate

HACKATHONS

MIT-iQuISE, iQuHACK-2024 | Python, Pytket

Feb. 2025

- Developed XXZ and Ising Hamiltonians, applied multi-order Trotterization using Quantinuum Pytket Framework

MIT-iQuISE, iQuHACK-2024 | Python, SciPy

Feb. 2024

- Built Quantum generative adversarial network (QGAN) model for photonic circuit using Quandela's Perceval library

TECHNICAL SKILLS

Languages: Python, Rust, C++, MySQL, JavaScript, HTML/CSS

Frameworks: OpenPulse, OpenQasm3, Qsharp, Qualtran, Cirq, Qiskit, openFermion, FastAPI

Developer Tools: Git, Bitbucket, Jira, Docker, VS Code, Visual Studio

Libraries: Qutip, PySCF, SciPy, Pandas, NumPy, Matplotlib